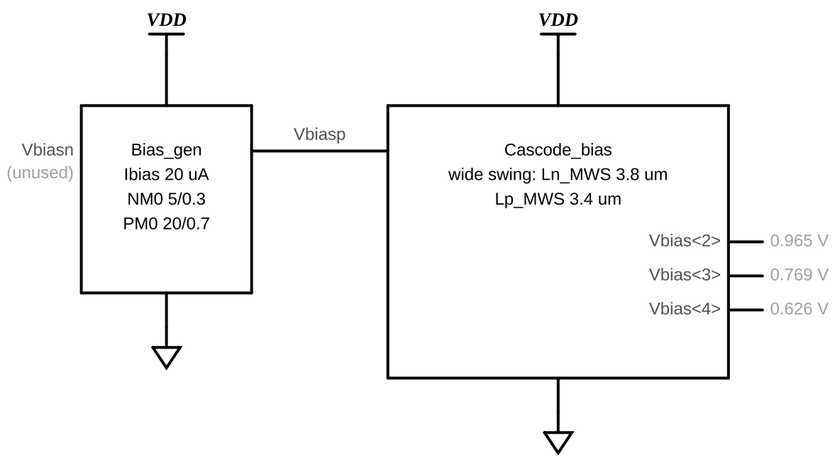
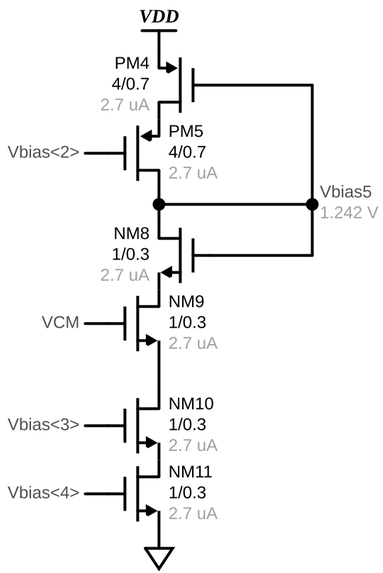
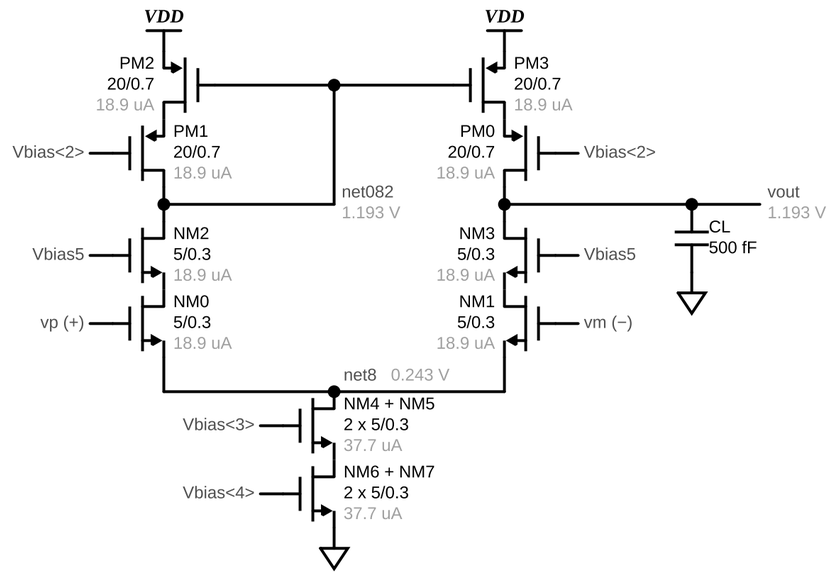
Telescopic_Amp: telescopic cascode amplifier with a bias replica for the NMOS cascodes



Bias replica





Sizes: multiplier x W/L in um (nf = 1), sky130 LVT devices. Currents and node voltages: tt, 27 °C, VDD = 1.8 V.
Replica branch: NM9 on VCM and the NM8 diode generate Vbias5 for the NMOS cascodes NM2 / NM3.
Simulated (ngspice, tt, 27 C): open-loop gain 51.3 dB, UGF 93.5 MHz.